

PROJECT: Houston Medical Center Expansion
SECTION 23 65 00 – COOLING TOWERS
ISSUED FOR BID – MOCK SPECIFICATION

PART 1 – GENERAL

1.01 SUMMARY

A. Provide factory-assembled, induced-draft, counterflow cooling towers as scheduled on the mechanical drawings.

B. Cooling towers shall serve the condenser water system for the central utility plant.

C. Work shall include cooling tower cells, fan assemblies, basin heaters, vibration switches, access ladders, safety cages, drift eliminators, distribution basins, supports, controls coordination, and manufacturer startup assistance.

1.02 RELATED DOCUMENTS

A. Mechanical Drawings: M601 Cooling Tower Schedule.

B. Division 23 General Mechanical Requirements.

C. Electrical coordination drawings.

D. Structural steel support drawings.

1.03 ACCEPTABLE MANUFACTURERS

A. Subject to compliance with the requirements of this specification, acceptable manufacturers are:

1. Marley
2. BAC
3. Evapco

B. Substitutions shall be submitted no later than ten calendar days before bid date.

C. Substitution requests shall include full performance data, dimensional data, electrical requirements, sound data, warranty terms, and written confirmation of CTI certification.

1.04 BASIS OF DESIGN

A. Basis of Design: Marley NC Series factory-assembled counterflow cooling tower.

B. Alternate manufacturers must demonstrate equal performance, footprint, sound level, accessories, and CTI certification.

1.05 PERFORMANCE REQUIREMENTS

A. Cooling towers shall meet the scheduled duty shown on Sheet M601.

B. Each cooling tower cell shall be capable of the following design conditions:

1. Flow Rate: 3,000 GPM
2. Entering Water Temperature: 95°F
3. Leaving Water Temperature: 85°F
4. Design Wet Bulb Temperature: 78°F
5. Cooling Range: 10°F
6. Service: Condenser water heat rejection

C. Cooling tower selection shall not exceed scheduled motor horsepower.

D. Cooling tower shall be selected for outdoor installation in Houston, Texas.

E. Performance shall be certified in accordance with CTI Standard 201.

1.06 QUALITY ASSURANCE

A. Cooling tower thermal performance shall be CTI certified.

B. Manufacturer shall have not less than ten years of experience manufacturing factory-assembled cooling towers.

C. Cooling tower shall comply with applicable local codes, energy codes, and project requirements.

D. Cooling tower shall be suitable for healthcare facility service where continuous operation and maintainability are required.

1.07 SUBMITTALS

A. Product data including dimensions, weights, materials, fan data, motor data, electrical requirements, sound data, drift rate, accessories, and maintenance clearances.

B. Cooling tower performance selection report.

C. CTI certification documentation.

D. Manufacturer installation instructions.

E. Manufacturer warranty.

F. Scheduled-vs-submitted comparison table.

G. Deviations and exclusions shall be clearly identified.

1.08 WARRANTY

A. Provide manufacturer standard warranty.

B. Warranty shall include defects in materials and workmanship for a minimum period of twelve months after startup or eighteen months after shipment, whichever occurs first.

PART 2 – PRODUCTS

2.01 COOLING TOWER TYPE

A. Factory-assembled, induced-draft, counterflow cooling tower.

B. Cooling tower shall be modular in construction and suitable for outdoor roof or grade-mounted installation.

2.02 MATERIALS OF CONSTRUCTION

A. Casing: Heavy-gauge galvanized steel or stainless steel where scheduled.

B. Cold Water Basin: Galvanized steel with corrosion-resistant coating.

C. Fan: Axial fan, statically and dynamically balanced.

D. Fill: PVC film fill suitable for condenser water applications.

E. Drift Eliminators: PVC drift eliminators with maximum drift rate of 0.0005% of circulating water flow.

F. Louvers: Manufacturer standard air inlet louvers.

2.03 FAN AND MOTOR REQUIREMENTS

A. Fan motor shall be premium efficiency, TEFC type.

B. Motor shall be suitable for variable frequency drive operation.

C. Electrical service shall be 480V, 3 phase, 60 Hz.

D. Fan motor horsepower shall not exceed 30 HP per cooling tower cell unless approved by engineer.

2.04 REQUIRED ACCESSORIES

- A. Variable frequency drive package.
- B. Basin sweeper piping.
- C. Electric basin heater with thermostat and low-water cutoff.
- D. Vibration switch.
- E. Access ladder with safety cage.
- F. External service platform where required by manufacturer maintenance clearances.
- G. Factory-installed float valve assembly.
- H. Make-up water connection.
- I. Overflow and drain connections.

2.05 SOUND REQUIREMENTS

- A. Cooling tower sound level shall not exceed 75 dBA at 50 feet under scheduled full-load operation.
- B. Sound data shall be submitted with proposal documentation when available.

2.06 STRUCTURAL AND ENVIRONMENTAL REQUIREMENTS

- A. Cooling tower shall be designed to withstand 130 MPH wind loading.
- B. Cooling tower shall be suitable for installation in a humid Gulf Coast environment.
- C. Unit shall be provided with corrosion-resistant materials and coatings suitable for outdoor service.

2.07 MANUFACTURER STARTUP

- A. Manufacturer representative shall provide startup assistance.
- B. Startup shall include inspection of installation, fan rotation verification, vibration check, basin heater verification, and control coordination.

PART 3 – EXECUTION

3.01 INSTALLATION

- A. Install cooling towers in accordance with manufacturer written instructions.

- B. Maintain manufacturer-required clearances for airflow, maintenance, access, and service.
- C. Coordinate final equipment dimensions and weights with structural support design.

3.02 FIELD QUALITY CONTROL

- A. Verify proper installation before startup.
- B. Confirm water flow, fan rotation, motor operation, vibration switch operation, and basin heater operation.
- C. Provide startup report to owner and engineer.

3.03 TRAINING

- A. Manufacturer representative shall provide owner training for cooling tower operation and maintenance.

3.04 CLOSEOUT DOCUMENTATION

- A. Provide final product data.
- B. Provide installation and operation manuals.
- C. Provide warranty documentation.
- D. Provide startup report.
- E. Provide recommended maintenance schedule.

END OF SECTION 23 65 00